

**GENERAL - ORGANIC - BIOCHEMISTRY**

Serial No.

002141

Prepared by the Examinations Institute of the American Chemical Society
Division of Chemical Education through its General - Organic - Biochemistry Committee.

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TO THE EXAMINER:

This test is designed to be taken with a special answer sheet on which the student records his or her responses. All answers are to be marked on this answer sheet, not in the test booklet. Each student should be provided with a test booklet, one answer sheet, and scratch paper; all of which must be turned in at the end of the examination period. The test is to be available to the students only during the examination period. You must collect and account for all exam booklets at the end of the examination. For complete instructions refer to the Directions for Administering Examinations. Only nonprogrammable calculators are permitted. **Digital cameras and cell phones are not permitted.** Norms are based on:

Subtest Part	I. General		II. Organic		III. Biochemistry	
	A	B	A	B	A	B
Number of items	40	20	40	20	40	20
Time limits, minutes	35	20	35	20	35	20

TO THE STUDENT:

DO NOT WRITE ANYTHING IN THIS BOOKLET! Do not turn this page until your instructor gives the signal to begin. A periodic table and other useful information is on the data sheet that is provided on page 2. When you are told to begin work, open the booklet and read the directions also on page 2.

DIRECTIONS
• When you have selected your answer, blacken the corresponding space on the answer sheet with a soft, black #2 pencil. Make a heavy, full mark, but no stray marks. If you decide to change an answer, erase the unwanted mark very carefully. • Make no marks in the test booklet. Do all calculations on scratch paper provided by your instructor. • There is only one correct answer to each question. Any questions for which more than one response has been blackened will not be counted. • Your score is based solely on the number of questions you answer correctly. It is to your advantage to answer every question. • Your test, answer sheet, and scratch paper must be turned in before leaving the examination room.

ABBREVIATIONS AND SYMBOLS			
amount of substance	n	enzyme	E
aqueous	aq	equilibrium constant	K
atmosphere	atm	formula molar mass	M
Avogadro constant	N_A	gram	g
Celsius temperature	$^{\circ}\text{C}$	joule	J
centi- (10^{-2})	c	kelvin	K
energy of activation	E_a	kilo- (10^3)	k
		liter	L
		measure of pressure	mmHg
		milli- (10^{-3})	m
		molar	M
		mole	mol
		pressure	P
		temperature, K	T
		volume	V

gas constant: $R = 0.0821 \text{ L} \cdot \text{atm} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

Note: Notation such as $\text{g} \cdot \text{mol}^{-1}$ is read as "grams per mole."

																		18																											
																		8A																											
1 1A H 1.008														2 2A He 4.003																															
3 Li 6.941	4 Be 9.012											13 3A B 10.81	14 4A C 12.01	15 5A N 14.01	16 6A O 16.00	17 7A F 19.00	18 Ne 20.18																												
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95																											
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80																												
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55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)																												
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 (269)	111 (272)	112 (277)	114 (277)																																	
<table border="1"> <tr> <td>58 Ce 140.1</td> <td>59 Pr 140.9</td> <td>60 Nd 144.2</td> <td>61 Pm (145)</td> <td>62 Sm 150.4</td> <td>63 Eu 152.0</td> <td>64 Gd 157.3</td> <td>65 Tb 158.9</td> <td>66 Dy 162.5</td> <td>67 Ho 164.9</td> <td>68 Er 167.3</td> <td>69 Tm 168.9</td> <td>70 Yb 173.0</td> <td>71 Lu 175.0</td> </tr> <tr> <td>90 Th 232.0</td> <td>91 Pa 231.0</td> <td>92 U 238.0</td> <td>93 Np (237)</td> <td>94 Pu (244)</td> <td>95 Am (243)</td> <td>96 Cm (247)</td> <td>97 Bk (247)</td> <td>98 Cf (251)</td> <td>99 Es (252)</td> <td>100 Fm (257)</td> <td>101 Md (258)</td> <td>102 No (259)</td> <td>103 Lr (262)</td> </tr> </table>																		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)
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GENERAL CHEMISTRY SUBTEST – PART A

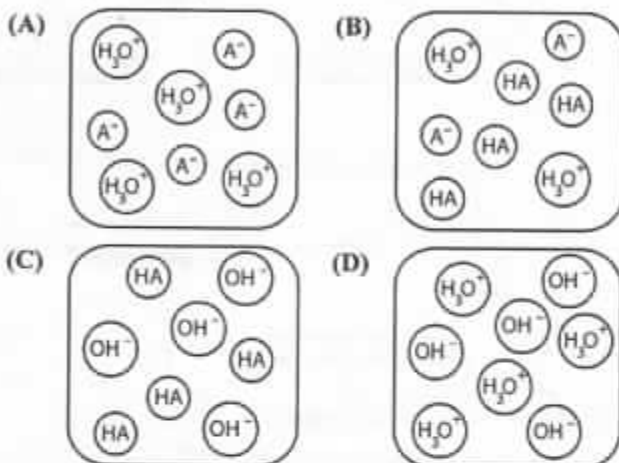
- What is the molar mass of magnesium hydroxide, $\text{Mg}(\text{OH})_2$, a common antacid?
(A) 58 g/mol (B) 56 g/mol
(C) 42 g/mol (D) 41 g/mol
- The average density of blood is 1.060 kg/L. The average human body contains 5.00 L of blood. What is the average mass of blood in the human body?
(A) 5.30 kg (B) 4.72 kg
(C) 3.94 kg (D) 0.212 kg
- The combustion of ethanol with oxygen forms carbon dioxide and water.
 $\text{CH}_3\text{CH}_2\text{OH}(\text{l}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
When this reaction is balanced the coefficients in order are:
(A) 1, 2, 3, 3 (B) 3, 3, 2, 1
(C) 1, 3, 2, 3 (D) 1, 3, 3, 2
- What is the approximate density of an average human?
(A) 120 g/mL (B) 1 g/mL
(C) 70 g/mL (D) 0.00006 g/mL
- How many moles of H atoms are in 0.50 moles of $\text{C}_6(\text{H}_{12}\text{O}_6)$, fructose?
(A) 0.50 moles H (B) 2.0 moles H
(C) 6.0 moles H (D) 12 moles H
- How many moles of $\text{Al}(\text{OH})_3$ are produced from the reaction of 6.0 moles of KOH according to the balanced equation:
 $\text{AlCl}_3 + 3\text{KOH} \rightarrow 3\text{KCl} + \text{Al}(\text{OH})_3$
(A) 18 moles (B) 6.0 moles
(C) 3.0 moles (D) 2.0 moles
- Which element is an alkali metal?
(A) Al (B) Fe (C) Mg (D) Rb
- J. J. Thompson used a cathode ray tube in order to discover the existence of what particle?
(A) electron (B) neutron
(C) photon (D) proton
- When nitrogen-14, $^{14}_7\text{N}$, is bombarded with helium-4, ^4_2He , a proton and one other particle are produced. What is the other particle? $^{14}_7\text{N} + ^4_2\text{He} \rightarrow ^1_1\text{H} + ?$
(A) $^{13}_6\text{C}$ (B) $^{17}_8\text{O}$ (C) $^{18}_9\text{F}$ (D) $^{19}_{10}\text{Ne}$
- Which mass is reported with 4 significant figures?
(A) 0.0040 g (B) 0.00400 g
(C) 0.004000 g (D) 4000 g
- Which pure substance will have the intermolecular force of hydrogen bonding with itself?
(A) CH_4 (B) HF (C) LiOH (D) CH_3F
- A laboratory experiment calls for you to make 1.20 L of a 0.100 M KNO_3 solution. You have a 5.00 M KNO_3 stock solution available. How many milliliters of the stock solution do you need to dilute to make the desired solution?
(A) 0.0240 mL (B) 16.7 mL
(C) 24.0 mL (D) 60.0 mL
- If 6.50 g of NaOH are dissolved in enough water to make 1750 mL of solution, what is the molarity of NaOH?
(A) 0.00371 M (B) 0.0929 M
(C) 0.163 M (D) 3.71 M
- Urine is considered normal when the pH range is 4.5 – 8.0. Test results show $[\text{H}^+] = 1 \times 10^{-9} \text{ M}$ for the urine of a patient. How does the pH of this patient's urine compare to the normal range?
(A) well below normal (B) below normal
(C) normal (D) above normal
- The chemical formula of sodium tungstate is Na_2WO_4 . What is the formula of calcium tungstate?
(A) CaWO_4 (B) $\text{Ca}(\text{WO}_4)_2$
(C) Ca_2WO_4 (D) $\text{Ca}_3(\text{WO}_4)_2$
- What is the systematic name for $\text{Fe}_2(\text{CO}_3)_3$?
(A) iron carbonate (B) di-iron tricarbonat
(C) iron (II) carbonate (D) iron (III) carbonate
- When pure liquid water is rapidly boiling, what are the bubbles made of?
(A) $\text{H}_2\text{O}(\text{g})$ (B) air
(C) $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ (D) $\text{CO}_2(\text{g})$

18. An amount of food is burned and releases 4000 calories of energy. What mass of water can be heated from 20.00 °C to 100.00 °C with this energy? The specific heat of water is 1.000 cal/g°C.
- (A) 0.02 g (B) 2 g
(C) 50 g (D) 300,000 g
19. You are asked to prepare 2 L of 5.5 % dextrose (m/v) solution for therapy. How much dextrose do you need to weigh out to prepare the solution?
- (A) 5.5 g (B) 55 g
(C) 550 g (D) 110 g
20. A sealed, rigid vessel contains 0.25 moles of Ne and 0.75 moles of O₂. The total pressure is 2.0 atm. What is the partial pressure of Ne?
- (A) 380 mm Hg (B) 760 mm Hg
(C) 1100 mm Hg (D) 1500 mm Hg
21. What is the formula of cobalt(III) sulfate?
- (A) Co₃SO₄ (B) Co₃(SO₄)₂
(C) Co₂(SO₄)₃ (D) CoSO₄
22. An insulin solution is given to a patient suffering from hyperglycemia at the rate of 20.0 mL/hr. The concentration of the insulin stock solution is 200. units/500. mL. How many units of insulin per hour is the patient is receiving?
- (A) 500 units/hr (B) 200 units/hr
(C) 10.0 units/hr (D) 8.00 units/hr
23. An isotope of selenium (Se) is used for pancreatic scans. What is the mass number of this isotope of selenium containing 41 neutrons?
- (A) 34 (B) 41 (C) 75 (D) 79
24. A(n) _____ bond forms when potassium and oxygen combine to form a compound.
- (A) hydrogen (B) ionic
(C) non-polar covalent (D) polar covalent
25. What species is the conjugate base of carbonic acid, H₂CO₃?
- (A) HCO₃⁻ (B) H₃CO₃⁺ (C) H₃O⁺ (D) CO₂
26. How many valence electrons does arsenic have?
- (A) 4 (B) 5 (C) 8 (D) 33
27. In a recent medical journal, a group of scientists claim, "We have found a plant extract that could cure diabetes." How would you classify this claim?
- (A) law (B) hypothesis
(C) serendipity (D) theory
28. Cr³⁺ ion contains a total of _____ electrons.
- (A) 3 (B) 27 (C) 24 (D) 21
29. Which physical state has a variable shape and cannot be readily compressed?
- (A) gas (B) liquid
(C) plasma (D) solid
30. The boiling point of liquid helium is 4K. Which calculation provides this temperature in °C?
- (A) 4 + 273 (B) 4 - 273
(C) 4 × (9/5) + 32 (D) (4 - 32) × (5/9)
31. How many moles of ibuprofen, C₁₃H₁₈O₂, are in a 250. mg dose of Motrin?
- | Formula weight, g/mol |
|--|
| C ₁₃ H ₁₈ O ₂ 206 |
- (A) 1.21 × 10⁻³ (B) 1.21 × 10⁻²
(C) 2.60 (D) 2.50 × 10²
32. Which substance can be described as a strong base?
- (A) CaCl₂ (B) HNO₃
(C) KOH (D) NH₃
33. Which property of water is unchanged when a small quantity of sodium chloride is added to it?
- (A) color (B) boiling point
(C) melting point (D) osmotic pressure
34. Normal saline is a solution of 0.90 % w/v of NaCl. If this often-used IV fluid were to be poured into a cup, what would the concentration of NaCl be?
- (A) less than 0.90 % w/v (B) exactly 0.90 % w/v
(C) more than 0.90 % w/v (D) cannot be determined
35. To prevent damage to blood cells, blood should be diluted in solutions that are _____.
- (A) apotonic (B) hypertonic
(C) hypotonic (D) isotonic
36. What is the charge of the cation in Fe₂O₃?
- (A) 3- (B) 2+ (C) 3+ (D) 6+

37. Which is a heterogeneous mixture?

- (A) salt water
(B) cranberry juice
(C) brewed tea with sugar
(D) hot chocolate with marshmallows

38. Which picture represents a strong acid (HA) dissolved in water (not shown)?



39. Which is an ionic compound?

- (A) SO_2 (B) CO_2 (C) SiO_2 (D) $\text{Ca}(\text{OH})_2$

40. Propane gas reacts with oxygen present in the air to form carbon dioxide and water. This reaction is an example of:
 $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g}) + \text{heat energy}$

- (A) combination (B) combustion
(C) decomposition (D) endothermic

End of Part A

Do not continue unless you have been instructed to do both Part A and Part B.

41. Which is the electron configuration for the element phosphorus?

- (A) $1s^2 2s^2 2p^6 2d^5$ (B) $1s^2 2s^2 2p^6 3s^2 3p^3$
(C) $1s^2 2s^2 3p^6 4s^2 5p^3$ (D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

42. What is the molecular shape of the OF_2 molecule?

- (A) bent (B) linear
(C) pyramidal (D) tetrahedral

43. According to its number of valence electrons, how many bonds will a chlorine atom typically form?

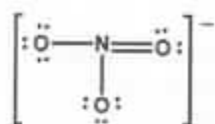
- (A) one (B) two (C) three (D) four

44. What substance is being oxidized in this single-replacement reaction?



- (A) Zn (B) $\text{Zn}(\text{NO}_3)_2$ (C) Ag (D) AgNO_3

45. How many total resonance structures are possible for the nitrate ion (NO_3^-)?

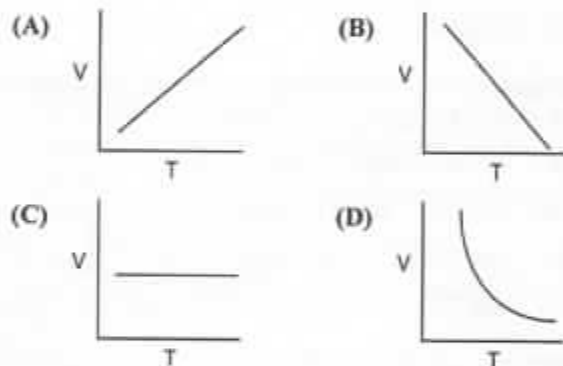


- (A) 1 (B) 2 (C) 3 (D) 4

46. Technetium-99, one of the most widely used radioisotopes in medicine, is employed in over 80% of all nuclear medicine procedures. This isotope has a half-life of 6 hours. If a patient is injected with a sample of technetium-99 that has an activity of 16 millicuries (mCi), when will the activity become 1.0 mCi?

- (A) 0.5 hours (B) 2.7 hours
(C) 12 hours (D) 24 hours

47. Which graph depicts the change in volume (V) with respect to a change in temperature (T) for one mole of a gas at constant pressure?



48. Which is *not* a correct Lewis structure?

- (A) $\text{H}-\ddot{\text{S}}-\text{H}$ (B) $\text{:C}\equiv\text{O:}$
(C) $\text{:O:}=\text{O}=\text{:O:}$ (D) all are correct Lewis structures

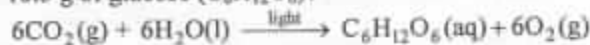
49. Which action can ensure that this reaction goes toward completion?
 $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{CO}_2(\text{g})$

- (A) add CO_2 (B) remove CO
(C) remove H_2 (D) remove H_2O

50. What is the concentration of NO_3^- in a 0.30 M $\text{Fe}(\text{NO}_3)_3$ solution?

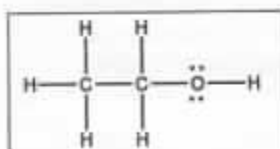
- (A) 0.10 M (B) 0.30 M
(C) 0.60 M (D) 0.90 M

51. In photosynthesis, as depicted by the reaction shown here, how many grams of CO_2 are required to produce 10.5 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)?



- (A) 5.83 (B) 15.4 (C) 63.0 (D) 180.

52. Which bond is the most polar in ethanol (shown here)?



- (A) C-C (B) C-O (C) C-H (D) O-H

53. "The nitrogen bends" is caused by decreasing pressure too quickly after a period of increased pressure (such as ascending too quickly after a dive). It gets its name from the fact that nitrogen bubbles may form in or near the joints, causing pain and causing the diver to "bend over". At a depth of 50. ft the pressure is 3.00 atm and the nitrogen bubble has a volume of 0.018 mL. Assuming constant temperature and number of moles, what will be the volume of the bubble at the sea level, where the pressure is 1.00 atm?

- (A) 0.054 mL (B) 3.0 mL
(C) 6.0 mL (D) 18 mL

54. What is the correct net ionic equation for the reaction of silver nitrate and sodium sulfate?

- (A) $\text{AgNO}_3 + \text{Na}_2\text{S} \rightarrow \text{Ag}_2\text{S} + 2\text{NaNO}_3$
(B) $2\text{AgNO}_3 + \text{Na}_2\text{SO}_4 \rightarrow \text{Ag}_2\text{SO}_4 + 2\text{NaNO}_3$
(C) $2\text{Ag}^+ + \text{SO}_4^{2-} \rightarrow \text{Ag}_2\text{SO}_4$
(D) $2\text{Ag}^+ + 2\text{NO}_3^- + 2\text{Na}^+ + \text{SO}_4^{2-} \rightarrow \text{Ag}_2\text{SO}_4 + 2\text{NaNO}_3$

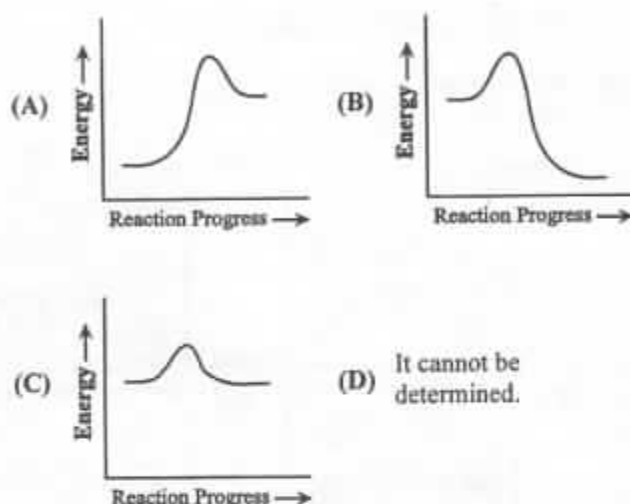
55. Which is a polar molecule?

- (A) (B)
(C) (D)

56. Which action will *not* increase the rate of a chemical reaction?

- (A) adding a catalyst to the reaction mixture
(B) doubling the size of the beaker or container in which the solutions are reacting.
(C) increasing the concentration of the reactant solutions.
(D) increasing the temperature at which the reaction occurs.

57. When a hot pack is activated, the solid inside dissolves in water and the resulting solution becomes warm. Which energy diagram best represents the process occurring inside the hot pack?



58. Which is the *strongest* acid?

- (A) hydrocyanic acid, $K_a = 4.9 \times 10^{-10}$
(B) carbonic acid, $K_a = 4.3 \times 10^{-7}$
(C) acetic acid, $K_a = 1.8 \times 10^{-5}$
(D) nitrous acid, $K_a = 4.5 \times 10^{-4}$

59. Before a shot is administered, the area is sterilized with an alcohol swab. The skin cools as the alcohol evaporates. This cooling sensation occurs because evaporation is an endothermic process in which

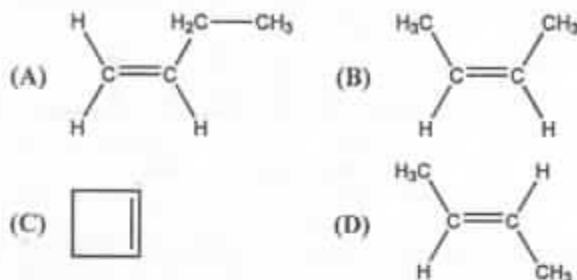
- (A) cold flows from the skin to the alcohol.
(B) cold flows from the alcohol to the skin.
(C) heat flows from the alcohol to the skin.
(D) heat flows from the skin to the alcohol.

60. When 5.6 grams of Fe is allowed to react with 5.0 g of elemental sulfur to form FeS , which reagent(s) is(are) used up completely and how much product is formed?
 $\text{Fe} + \text{S} \rightarrow \text{FeS}$

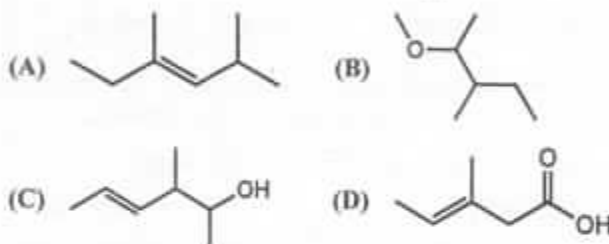
- (A) Fe is used up completely, and 8.8 g of FeS is formed.
(B) S is used up completely, and 8.7 g of FeS is formed.
(C) Fe is used up completely, and 7.2 g of FeS is formed.
(D) S is used up completely, and 7.8 g of FeS is formed.

**ORGANIC CHEMISTRY SUBTEST
PART A**

61. Which structure is *trans*-2-butene?



62. Which molecule has the lowest boiling point?

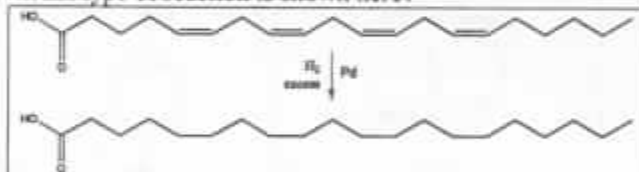


63. In the liver, the enzyme alcohol dehydrogenase (ADH) converts ethanol into ethanal, as show in the equation below. This process is known as _____.



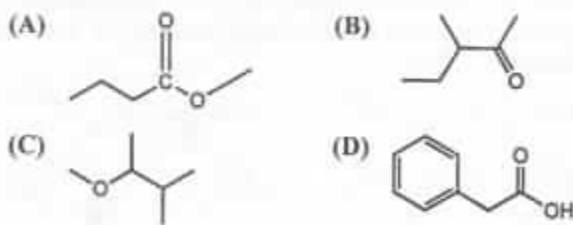
- (A) oxidation (B) reduction
(C) hydrogenation (D) dehydration

64. What type of reaction is shown here?

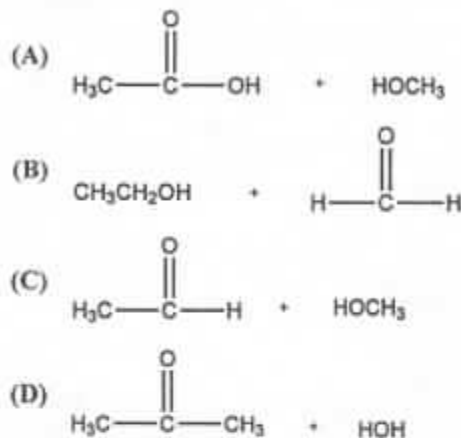
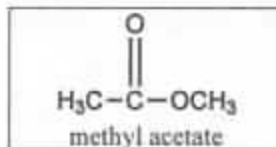


- (A) halogenation (B) hydration
(C) hydrogenation (D) hydrohalogenation

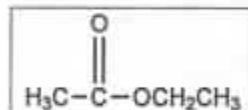
65. Which structure is an ether?



66. What two reactants can be used to synthesize methyl acetate with acid catalyst?

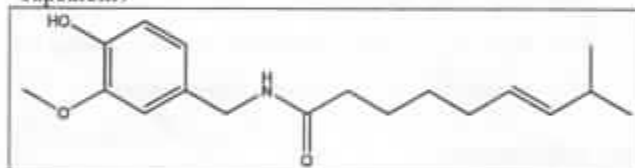


67. This molecule is found in nail polish remover. What is the functional group?



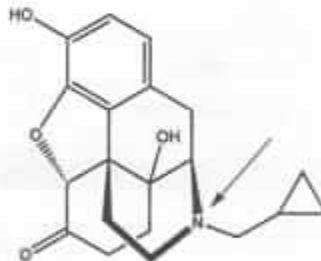
- (A) carboxylic acid (B) ester
(C) ether (D) ketone

68. Capsaicin is the main active compound found in peppers that is responsible for the hotness of peppers such as chili peppers. What functional group is *not* present in capsaicin?



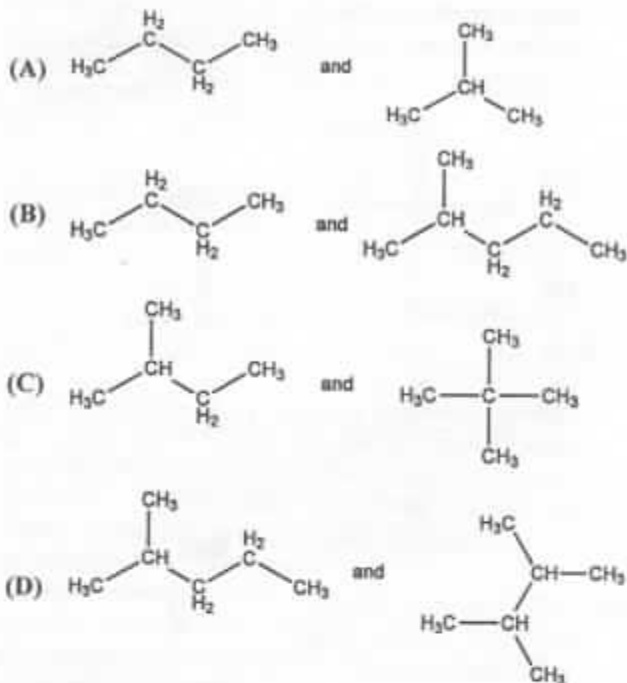
- (A) phenol (B) amide
(C) amine (D) ether

69. Naltrexone, shown here, is an inhibitor used in the management of alcohol and opiate dependence. The functional group indicated by the arrow is classified as a _____ amine.



- (A) primary (B) secondary
(C) tertiary (D) quaternary

70. Which pair of molecules are *not* isomers?

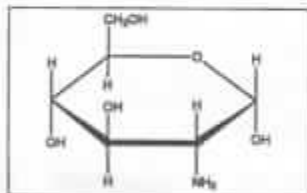


71. What is the difference between an unsaturated and saturated hydrocarbon?

Unsaturated hydrocarbons contain: **Saturated hydrocarbons contain:**

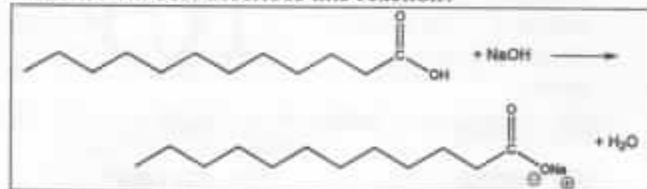
- (A) no carboxylic acid carboxylic acid
(B) only *trans* double bonds only *cis* double bonds
(C) even number of carbons at least one double bond
(D) at least one double bond single bonds

72. Glucosamine, a compound found in shellfish, is an example of which functional group?



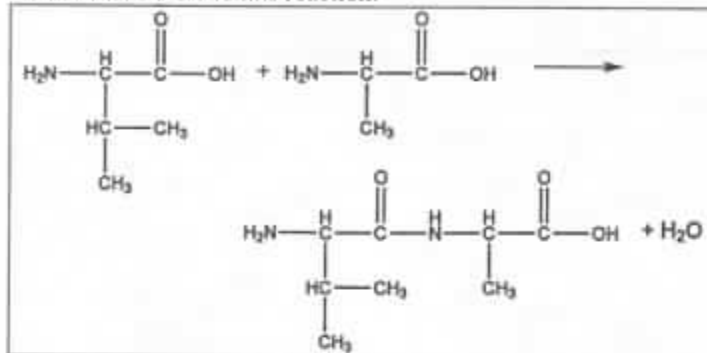
- (A) acetal (B) hemiacetal
(C) aldehyde (D) ester

73. Which term best describes this reaction?



- (A) dehydration (B) hydration
(C) hydrolysis (D) neutralization

Items 74-75 refer to this reaction:



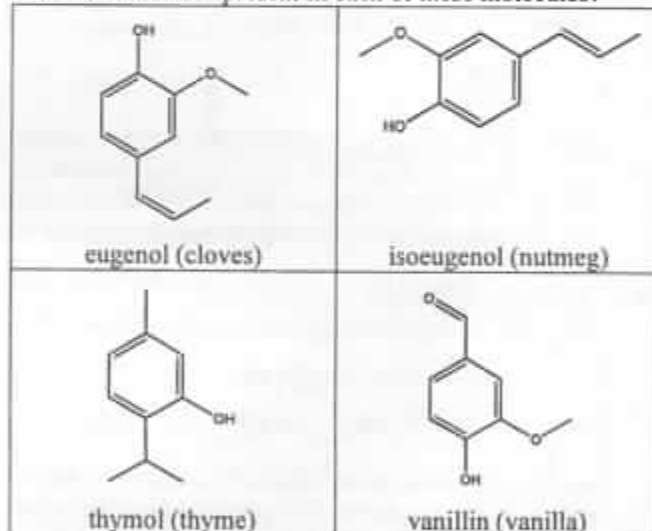
74. What type of reaction is this?

- (A) condensation (B) elimination
(C) hydration (D) hydrolysis

75. What functional group is formed when the two reactant molecules combine to form the final product?

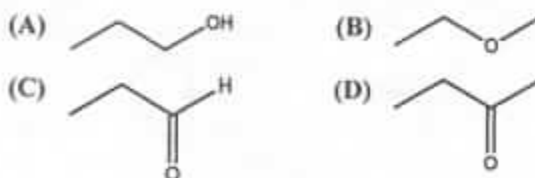
- (A) aldehyde (B) amide
(C) amine (D) carboxylic acid

76. Several essential oils of plants which produce the odor or flavor of the plants are shown. What is the common structural feature present in each of these molecules?

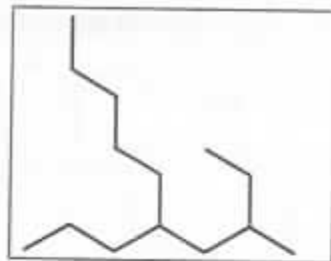


- (A) cyclohexane ring (B) ester
(C) ether (D) phenol

77. Which compound exhibits hydrogen bonding in a pure sample?

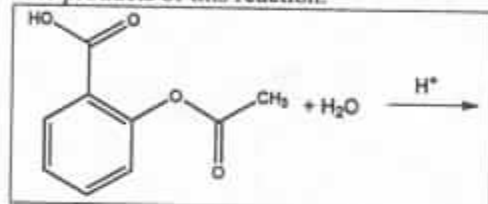


78. When named following IUPAC rules, what is the parent chain?



- (A) heptane (B) octane
(C) nonane (D) decane

79. The hydrolysis of aspirin occurs in the stomach. Predict the products of this reaction.



- (A)
- (B)
- (C)
- (D)

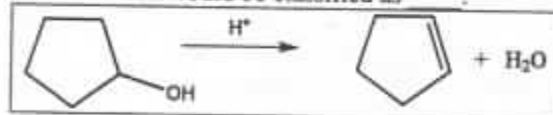
80. Which molecule can be classified as an organic base?

- (A)
- (B)
- (C)
- (D)

81. Which combination of the reactants results in the formation of a hemiacetal?

- (A) aldehyde and alcohol under acidic conditions
(B) aldehyde and amine under acidic conditions
(C) aldehyde and ketone under acidic conditions
(D) aldehyde and aldehyde under acidic conditions

82. This reaction would be classified as

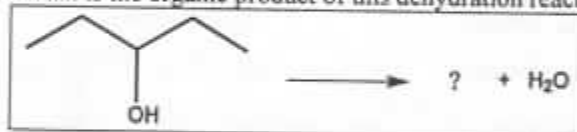


- (A) reduction (B) dehydroxylation
(C) dehydration (D) condensation

83. Which compound is most soluble in water?

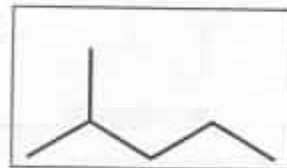
- (A)
- (B)
- (C)
- (D)

84. What is the organic product of this dehydration reaction?



- (A)
- (B)
- (C)
- (D)

85. What is the correct formula for this structure?

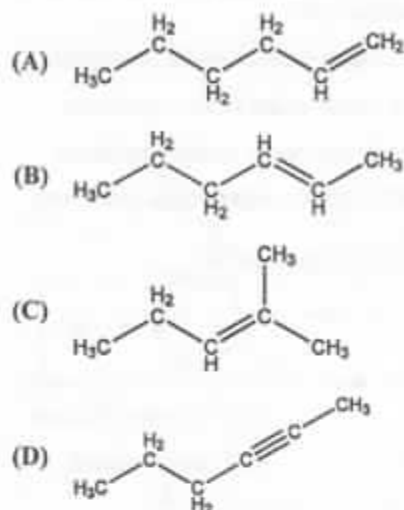


- (A) C_5H_{10} (B) C_5H_{12} (C) C_6H_{13} (D) C_6H_{14}

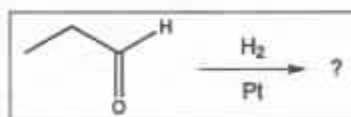
86. Which type of molecule will give a positive result with Tollens' reagent, an oxidizing agent?

- (A) aldehyde (B) carboxylic acid
(C) ester (D) ketone

87. Which molecule could exist as *cis-trans*- isomers?



88. What is the product of this reaction?

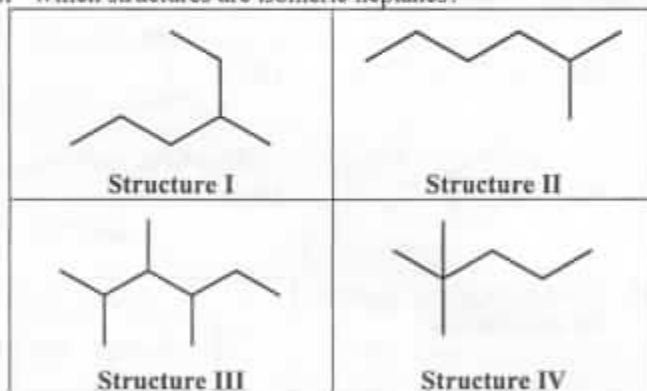


- (A) propane (B) propanal
(C) 1-propanol (D) 2-propanone

89. What type of alcohol can undergo mild oxidation to yield an aldehyde?

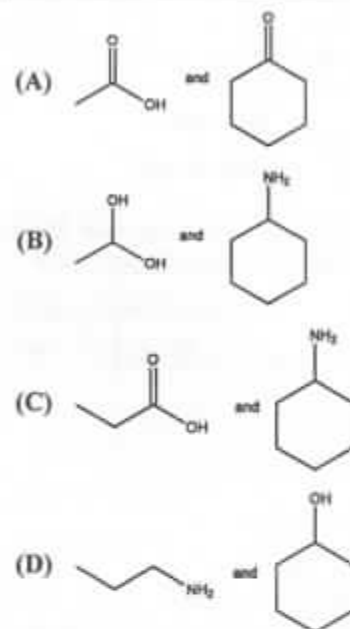
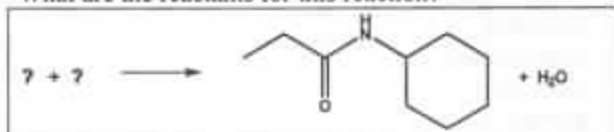
- (A) primary (B) secondary
(C) tertiary (D) all are possible

90. Which structures are isomeric heptanes?

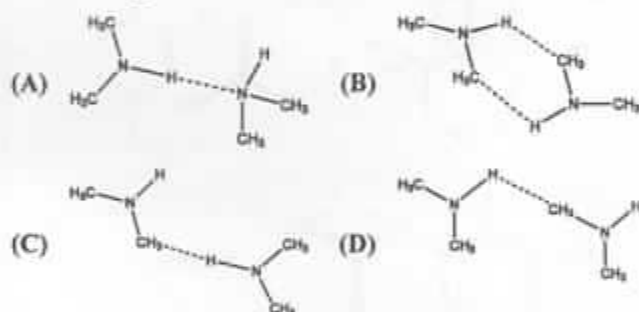


- (A) II and IV only (B) III and IV only
(C) I, II and III only (D) I, II and IV only

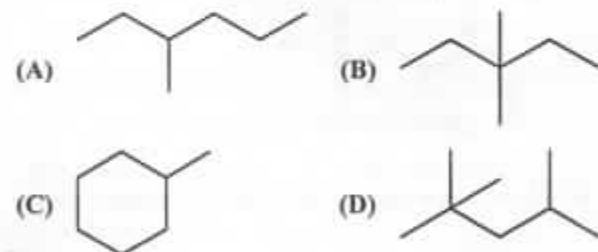
91. What are the reactants for this reaction?



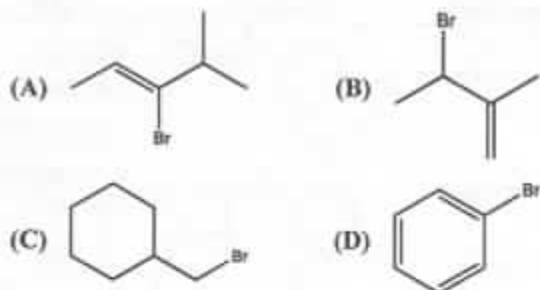
92. Which depiction best shows hydrogen bonding between two dimethylamine molecules?



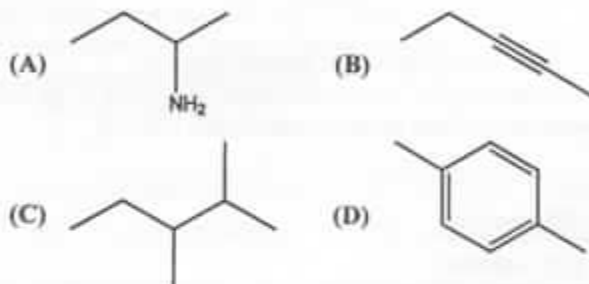
93. Which molecule does *not* have a tertiary carbon atom?



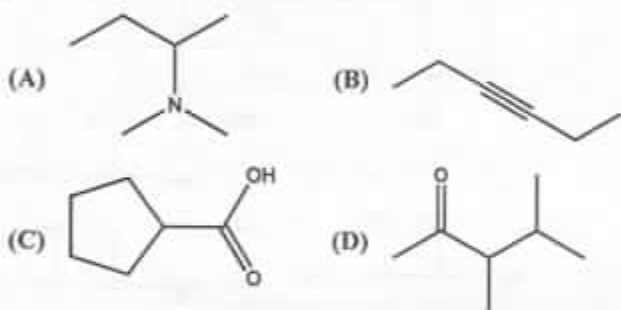
94. Which structure corresponds to the molecular formula $C_6H_{11}Br$?



95. All of the structures are non-polar *except*



96. Which compound is most soluble in a non-polar solvent, like hexane?



97. According to VSEPR theory, what is the geometry of a carbon atom in an alkene bond?



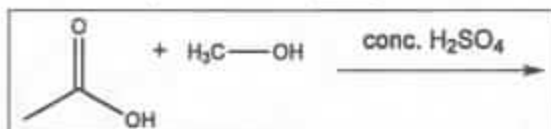
98. Which of these compounds contains the most hydrogen atoms?



99. Many of the "fruity-smelling" compounds are due to



100. The reaction between a carboxylic acid, an alcohol and an acid catalyst is known as ____.



- (A) esterification (B) neutralization
(C) reduction (D) saponification

End of Part A

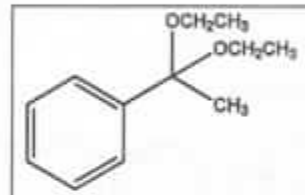
Do not continue unless you have been instructed to do both Part A and Part B.

101. Which reagent is likely to react with 1-pentanamine?



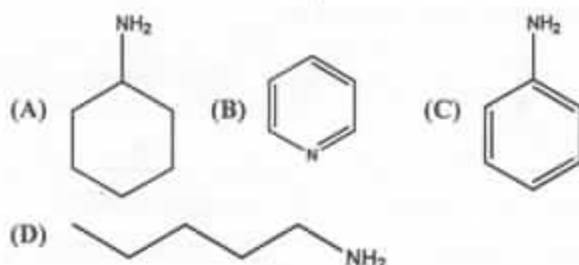
- (A) NaOH(aq) (B) NaHCO₃(aq)
(C) NaCl(aq) (D) HCl(aq)

102. Which combination of products will be produced from this acetal when it is hydrolyzed?

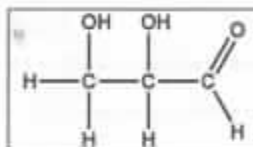


- (A) and CH₃OH
(B) and CH₃CH₂OH
(C) and
(D) and CH₃CH₂OH

103. Which molecule is a heterocyclic amine?

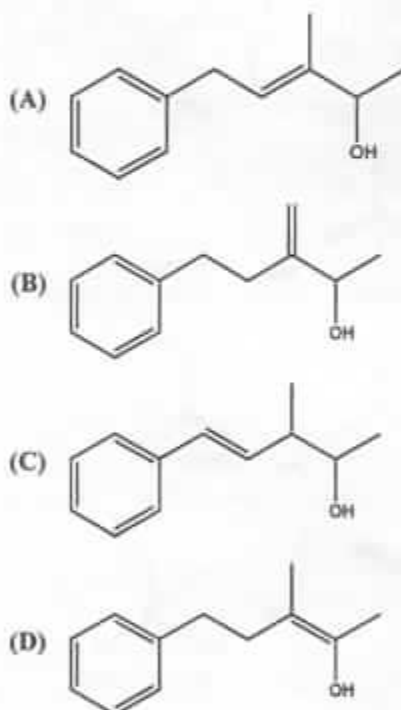
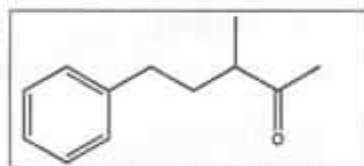


104. In this molecule, how many of the carbon atoms are chiral?

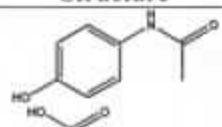
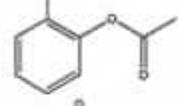


- (A) 0 (B) 1 (C) 2 (D) 3

105. Which structure is the enol tautomer of this ketone?



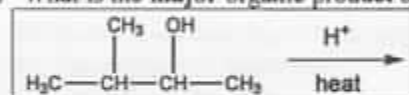
106. Use information in this table to answer the question.

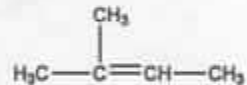
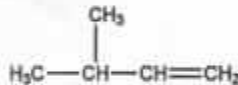
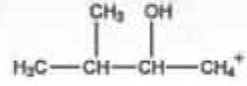
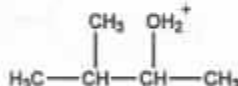
Active Ingredient	Structure	Solubility in water
acetaminophen		12.8 mg/mL at 20 °C
aspirin		0.30 mg/mL at 20 °C
caffeine		2.2 mg/mL at 20 °C

Of the active ingredients in pain relievers (acetaminophen, aspirin and caffeine), which is the least soluble in water at room temperature (20 °C)?

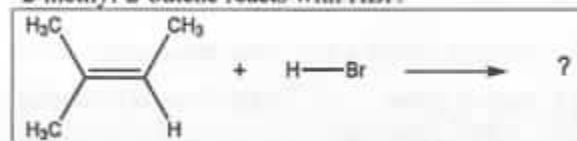
- (A) acetaminophen
 (B) aspirin
 (C) caffeine
 (D) all would be soluble with added CH_2Cl_2

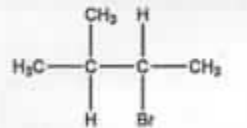
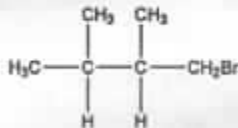
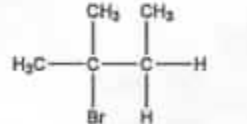
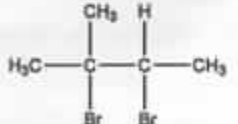
107. What is the major organic product of this reaction?



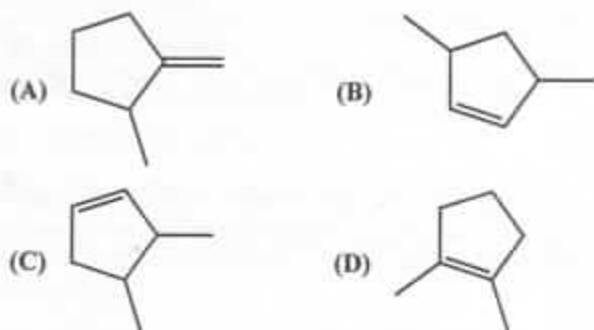
- (A)  (B) 
 (C)  (D) 

108. Which molecule is the major product when 2-methyl-2-butene reacts with HBr?

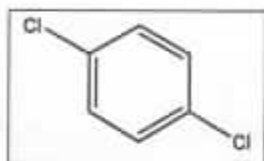


- (A)  (B) 
 (C)  (D) 

109. Which substance will **not** form 1,2-dimethylcyclopentane upon catalytic hydrogenation with H_2/Ni ?

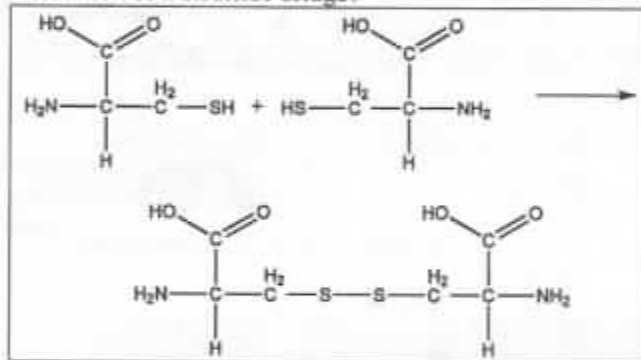


110. What is the name of the structure when two chlorines are attached to a benzene molecule as shown?



- (A) 1,3-chlorobenzene (B) *m*-dichlorobenzene
(C) *o*-dichlorobenzene (D) *p*-dichlorobenzene

111. What is the word that best describes this reaction: the formation of a disulfide bridge?

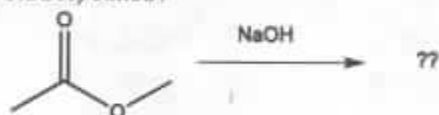


- (A) condensation (B) hydrolysis
(C) oxidation (D) reduction

112. What functional group can be formed when a ketone undergoes reduction?

- (A) primary alcohol (B) secondary alcohol
(C) aldehyde (D) carboxylic acid

113. What is the reaction with an ester and a strong base (like NaOH) called?

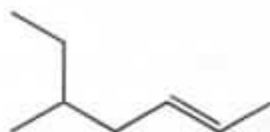


- (A) esterification (B) neutralization
(C) reduction (D) saponification

114. What is the functional group in CH_3COCH_3 ?

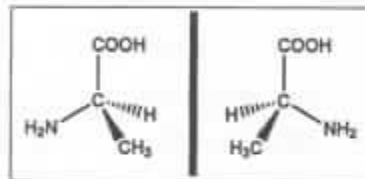
- (A) aldehyde (B) ester
(C) ether (D) ketone

115. What is the correct IUPAC name of this compound?



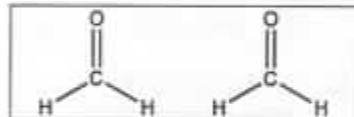
- (A) *cis*-3-methyl-5-heptene
(B) *trans*-2-methyl-5-hexene
(C) *trans*-5-methyl-2-heptene
(D) *trans*-1-ethyl-1-methyl-2-butene

116. What is the relationship between the two molecules of alanine, an amino acid, shown here?



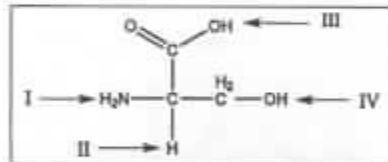
- (A) conformers (B) diastereomers
(C) enantiomers (D) geometric isomers

117. What intermolecular forces occur between these two molecules?



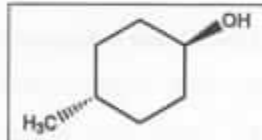
- (A) London dispersion only
(B) London dispersion and dipole-dipole interactions
(C) London dispersion and hydrogen bonding
(D) London dispersion, hydrogen bonding, dipole-dipole

118. Which of the hydrogen atoms indicated in the structure is the most acidic?



- (A) I (B) II (C) III (D) IV

119. What product is formed by the acid-catalyzed dehydration of *trans*-4-methylcyclohexanol?



- (A) 4-methylcyclohexane (B) 4-methylcyclohexene
(C) 5-methylcyclohex-3-ene (D) cyclohexanol

120. According to structural theory, how many bonds is a bromine atom allowed to form?

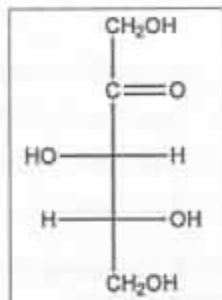
- (A) one (B) two (C) three (D) four

**BIOCHEMISTRY SUBTEST
PART A**

121. What type of biomolecule contains an ester?

- (A) amino acid (B) carbohydrate
(C) protein (D) triglyceride

122. The Fisher projection of D-ribulose is shown. This molecule is classified as a(n) ____.

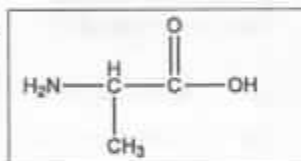


- (A) aldopentose (B) aldohexose
(C) ketopentose (D) ketohexose

123. Amylose is a natural polymer. What is the monomer of amylose?

- (A) D-ribose (B) D-glucose
(C) D-fructose (D) amylase

124. To the right is the structure of the amino acid, alanine. At physiological pH, what is the actual structure of alanine?

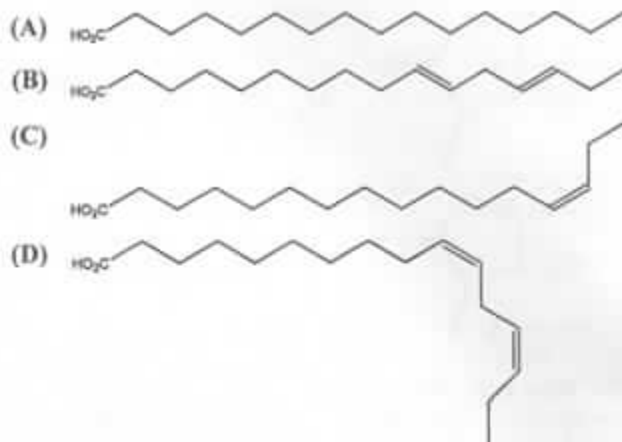


- (A) (B)
- (C) (D)

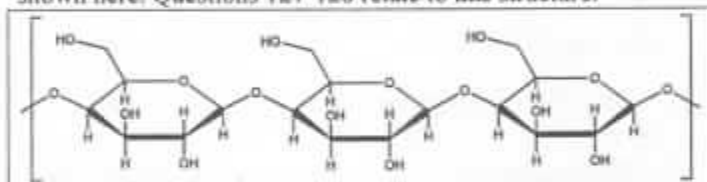
125. Which is *not* a characteristic of most natural fatty acids?

- (A) Fatty acids can have more than one C=C.
(B) Unsaturated fatty acids have all *cis*- double bonds.
(C) Fatty acids have an even number of carbon atoms.
(D) Unsaturated fatty acids have all *trans*- double bonds.

126. Which fatty acid is likely to have the lowest melting point?



Beta-glucans is listed as an ingredient in a supplement in an immune booster. The structure of an example beta-glucan is shown here. Questions 127-128 relate to this structure.



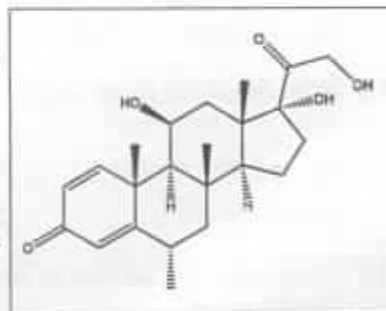
127. To what class of biomolecules does this molecule belong?

- (A) carbohydrates (B) enzymes
(C) lipids (D) proteins

128. What kind of linkage connects monomers in this polymer?

- (A) α 1 $\rightarrow$ 3 (B) α 1 $\rightarrow$ 4
(C) β 1 $\rightarrow$ 3 (D) β 1 $\rightarrow$ 4

129. Medrol (methylprednisolone) prevents the release of substances in the body that cause inflammation and is used to treat arthritis. To what general class of molecules does Medrol belong?



- (A) eicosanoids (B) opioids
(C) prostaglandins (D) steroids

130. The primary structure of a protein depends solely upon ____.

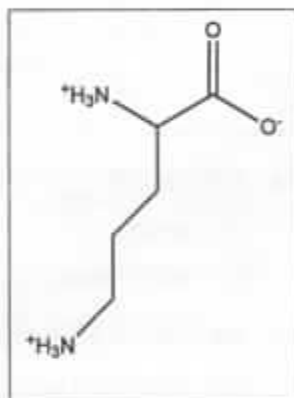
- (A) covalent bonds
- (B) disulfide linkage
- (C) electrostatic interactions
- (D) hydrophobic interaction

131. Sickle cell anemia is a blood disorder resulting from a change in the amino acid sequence of the hemoglobin protein. At what structural level of a protein does this change occur?

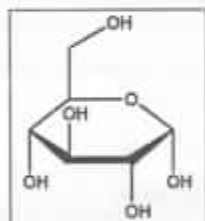
- (A) primary
- (B) secondary
- (C) tertiary
- (D) quarternary

132. The amino acid L-ornithine plays a central role in the urea cycle, and its structure is shown. How is the amino acid, ornithine, classified?

- (A) hydrophobic non-polar
- (B) hydrophobic polar
- (C) hydrophilic acidic
- (D) hydrophilic basic



133. Which structure is the anomer of the sugar shown here?



- (A)
- (B)
- (C)
- (D)

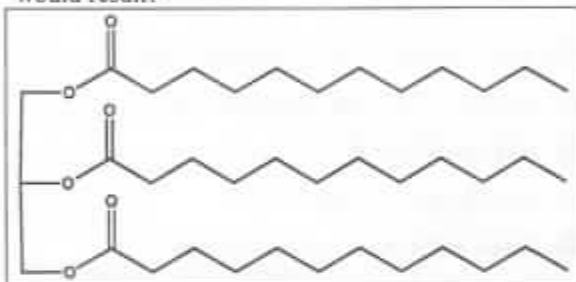
134. Which molecule is an example of an omega-3 fatty acid?

- (A)
- (B)
- (C)
- (D)

135. In the body, a ____ condition may result when glucose levels drop too low and is generally seen only in people who have diabetes.

- (A) hyperglycemic
- (B) hypoglycemic
- (C) hypertonic
- (D) hypotonic

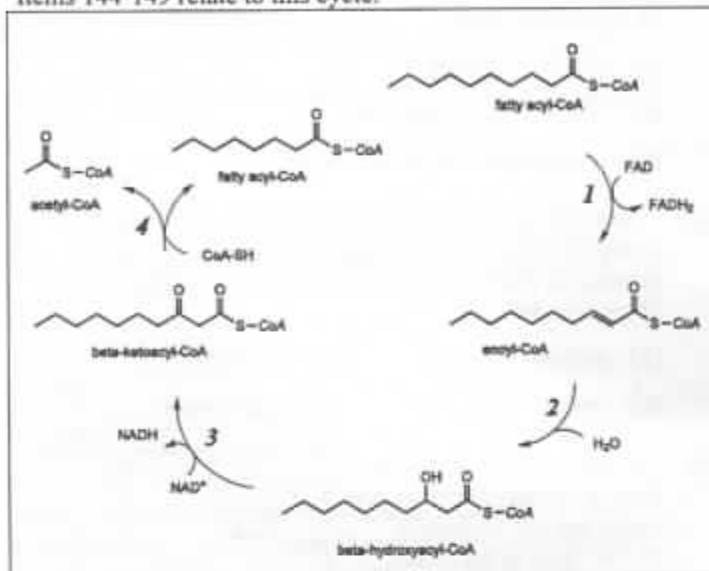
136. Consider this structure of a triacylglycerol. If this compound were hydrolyzed by a lipase, what product(s) would result?



- (A)
- (B)
- (C)
- (D)

137. The peptide bond is an example of which organic functional group?
- (A) amide (B) amine
(C) ester (D) ether
138. Which biomolecule would be a dietary source of nitrogen?
- (A) carbohydrates (B) lipids
(C) nucleic acids (D) proteins
139. Which biomolecule is most likely to dissolve in water?
- (A) carbohydrates (B) lipids
(C) nucleic acids (D) proteins
140. Given equal amounts, which type of food will provide the most energy?
- (A) high carbohydrate (B) high fat
(C) high electrolyte (D) high protein
141. Which is an example of an oxidation reaction?
- (A) $\text{alcohol} + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{alkene} + \text{H}_2\text{O}$
(B) $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$
(C) $\text{FADH}_2 \rightarrow \text{FAD} + 2\text{H}^+ + 2\text{e}^-$
(D) $\text{NAD}^+ + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{NADH} + \text{H}^+$
142. Which statement about DNA and RNA is *not* true?
- (A) They have four nitrogenous bases.
(B) They are only found in the nucleus.
(C) They are polymers made of nucleotides.
(D) They have a sugar-phosphate backbone.
143. Protein-based drugs are in development for the treatment of many diseases. The drawback to these drugs is that they are given to patients as either IV or injections. Why is drug delivery via IV or injection and not in pill form?
- (A) Protein-based drugs cannot be made into pills.
(B) The IV or injection drug is cheaper than a pill.
(C) Protein-based drugs taken orally will denature in the stomach acid.
(D) Injection into the blood stream delivers the protein-based drug faster

Use the reaction cycle shown here for the next six questions. Items 144-149 relate to this cycle.

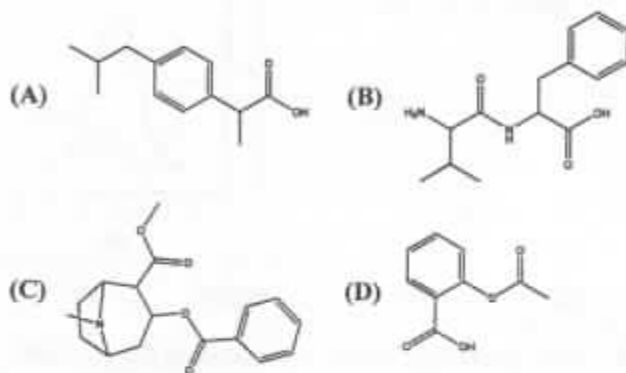


144. This cycle is _____ because it breaks down molecules.
- (A) aerobic (B) anabolic
(C) anaerobic (D) catabolic
145. What type of reaction is step 2 in the cycle?
- (A) reduction (B) oxidation
(C) hydrolysis (D) hydration
146. What type of reaction does the acyl chain undergo in step 3 in the cycle?
- (A) reduction (B) oxidation
(C) hydrolysis (D) hydration
147. If a fatty acyl-CoA were sixteen carbons in length, how many beta-oxidation cycles will it have to undergo?
- (A) 4 (B) 7 (C) 12 (D) 16
148. How many NADH would be generated from the breakdown of fatty acyl-CoA that is 10 carbons in length?
- (A) 4 (B) 5 (C) 9 (D) 10
149. The flow-through of this cycle would be _____ when blood sugar is low.
- (A) high (B) low
(C) unchanged (D) unknown
150. Which is *not* a coenzyme?
- (A) CoA (B) FAD (C) H_2O (D) NAD^+

151. Alpha helices and beta-pleated sheets are examples of the _____ structure of proteins.

- (A) primary (B) tertiary
(C) secondary (D) quaternary

152. Which structure contains a peptide bond?



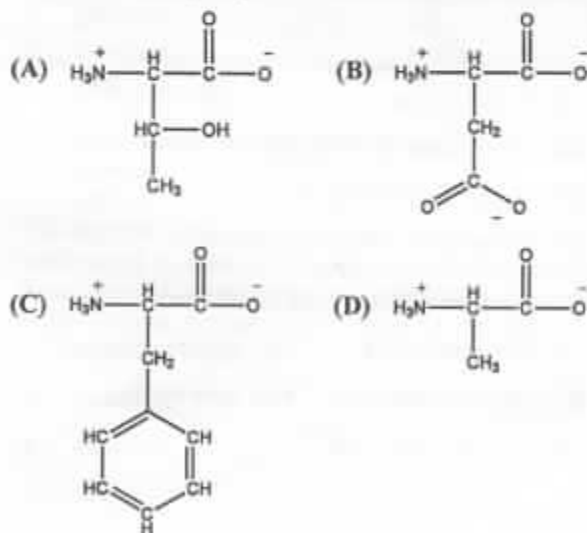
153. Niemann-Pick disease is caused by low levels of the enzyme sphingomyelinase. Which class of biomolecule does this enzyme act upon?

- (A) lipids (B) proteins
(C) nucleic acids (D) carbohydrates

154. When lysine ($R = -(CH_2)_4NH_3^+$) and glutamate ($R = -(CH_2)_2CO_2^-$) are found near to each other within a protein, they may stabilize the structure through _____.

- (A) covalent interaction
(B) disulfide linkage
(C) electrostatic interaction
(D) hydrophobic interaction

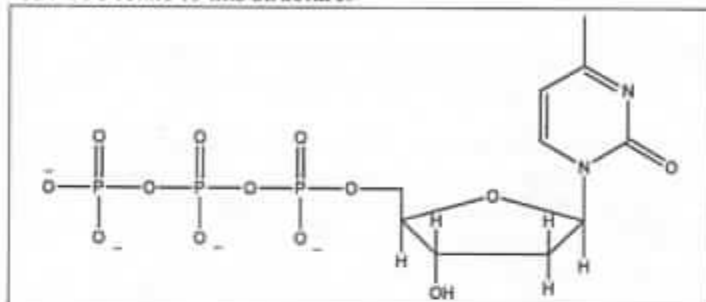
155. Which is most hydrophobic?



156. People suffering from lactose intolerance are deficient in what compound?

- (A) galactose (B) lactase
(C) lactose (D) maltase

Use the structure shown here for the next two questions. Items 157-158 relate to this structure.



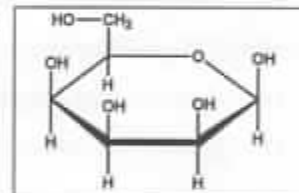
157. What sugar connects the nitrogenous base to the phosphates?

- (A) deoxyribose (B) ribose
(C) deoxyglucose (D) glucose

158. What type of nitrogenous base is present?

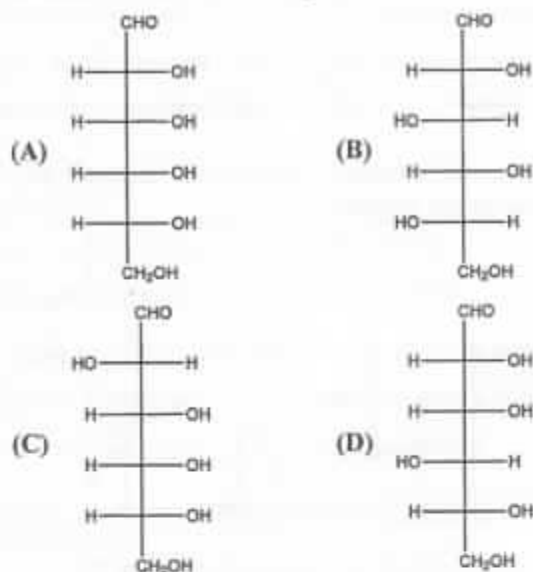
- (A) piperidine (B) purine
(C) pyridine (D) pyrimidine

159. How many of the hydrogen atoms shown in this molecule could participate in hydrogen bonding with the oxygen atom in water?



- (A) 1 (B) 4 (C) 5 (D) 12

160. Which molecule is an L-sugar?



End of Part A

Do not continue unless you have been instructed to do both Part A and Part B.

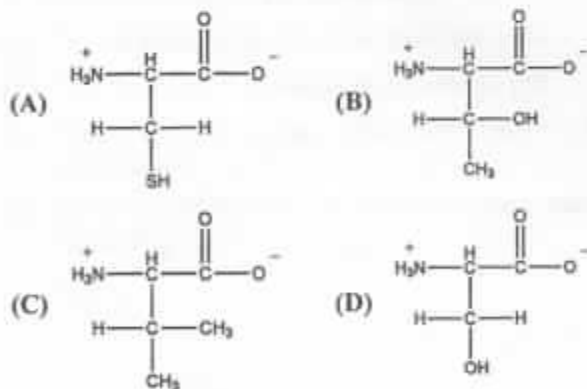
161. Which pair of molecules could be called an anomeric pair?

- (A) α -D-galactose and α -L-galactose
 (B) β -D-galactose and β -D-glucose
 (C) α -D-galactose and β -D-galactose
 (D) β -L-fructose and α -L-ribose

162. Retroviruses, such as HIV, have _____ as their genetic material.

- (A) carbohydrates (B) DNA
 (C) proteins (D) RNA

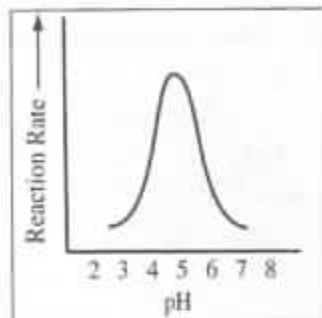
163. Which amino acid has more than one chirality center?



164. DFP inhibits acetylcholine esterase by remaining bound to the active site. How is this inhibitor classified?

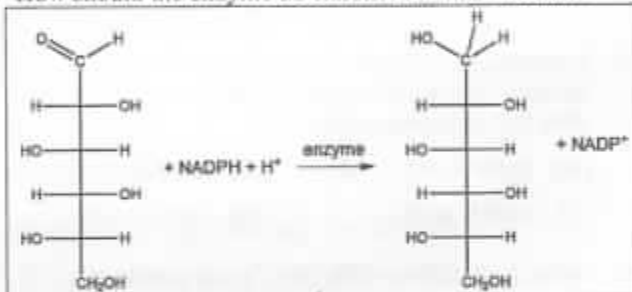
- (A) competitive (B) non-competitive
 (C) irreversible (D) reversible

165. Which pH is optimal for the enzyme reaction?



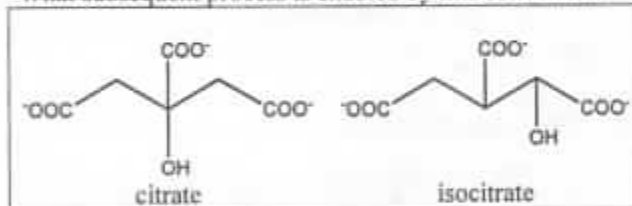
- (A) 4 (B) 5 (C) 6 (D) 7

166. How should the enzyme be classified in this reaction?



- (A) hydrolase (B) isomerase
 (C) oxidoreductase (D) transferase

167. In the citric acid cycle, citrate is converted to isocitrate. What subsequent process is enabled by this conversion?



- (A) hydrolysis of acetyl CoA
 (B) dehydration the molecule
 (C) oxidation of the alcohol
 (D) conversion of NADH to NAD⁺

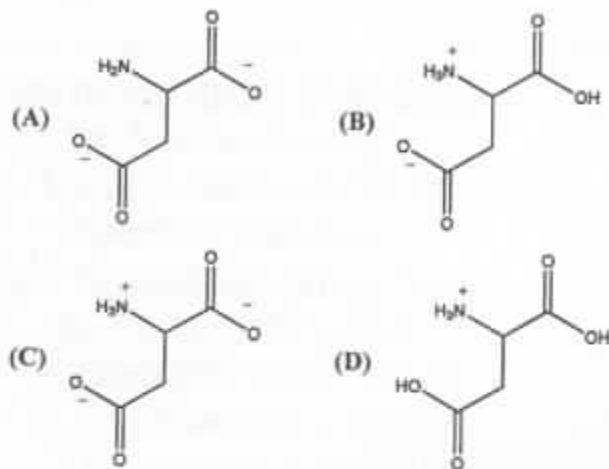
168. What process oxidizes NADH and FADH₂?

- (A) citric acid cycle (B) electron transport
 (C) fatty acid oxidation (D) urea cycle

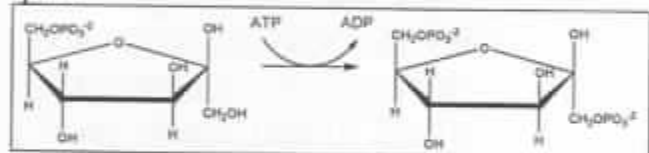
169. Which process does *not* occur in the nucleus?

- (A) integration (B) replication
(C) transcription (D) translation

170. Which structure represents the predominant form of aspartic acid at pH 12.0?



171. In this transformation, what type of reaction is taking place?



- (A) hydration (B) isomerization
(C) oxidoreduction (D) phosphorylation

172. Gluconeogenesis is a:

- (A) biochemical pathway that converts glucose from glycogen.
(B) biochemical pathway that converts glucose into glycogen.
(C) biochemical pathway that produces glucose from pyruvate.
(D) biochemical pathway that converts glucose into pyruvate.

173. The digestive enzyme chymotrypsin is initially synthesized as chymotrypsinogen to protect the pancreas from digestion. Upon cleavage by trypsin, it becomes the active enzyme chymotrypsin. This is an example of _____ enzyme regulation.

- (A) allosteric (B) genetic modulation
(C) isozyme (D) zymogen (proenzyme)

174. Cholesterol:

- (A) is a steroid that is responsible for increasing the fluidity of lipid bilayers.
(B) is a rigid, fused-ring steroid that can decrease the fluidity of biological membranes.
(C) can disrupt stacking in a lipid bilayer and thus increase the transition temperature.
(D) is found in biological membranes because it is used to transport large, polar molecules into a cell.

175. The electron transport chain is responsible for shuttling protons across the inner mitochondrial membrane to create a proton gradient. The potential energy of this gradient is then used to drive the synthesis of ATP. Oxidation of which coenzyme would provide more energy for ATP synthesis?

- (A) NAD^+ (B) NADH (C) FAD (D) FADH_2

176. What would be the complementary RNA strand to 5'-GCTAGGC-3'?

- (A) 3'-CGAUCCG-5' (B) 3'-GCCUAGC-5'
(C) 3'-CGATCCG-5' (D) 3'-GCCTAGC-5'

177. During protein translation, the _____ codon base pairs with the _____ anticodon.

- (A) DNA, RNA (B) RNA, DNA
(C) mRNA, tRNA (D) tRNA, mRNA

178. A codon binds to its anticodon with:

- (A) amino-acyl bonds (B) ester bonds
(C) hydrogen bonds (D) peptide bonds

179. In which process would protein molecules lose their biological function?

- (A) amidization (B) denaturation
(C) esterification (D) saponification

180. NADH produced during glycolysis under anaerobic conditions is regenerated to NAD^+ by conversion of:

- (A) pyruvate $\rightarrow$ lactate
(B) glucose $\rightarrow$ glucose-6-phosphate
(C) phosphoenolpyruvate $\rightarrow$ pyruvate
(D) glyceraldehyde-3-phosphate $\rightarrow$ 1,3-bisphosphoglycerate

END OF TEST